

SHORT COMMUNICATION

Radioprotective Action of Extracellular Adenosine on Bone Marrow Cells in Mice Exposed to Gamma Rays as Assayed by the Micronucleus Test

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The frequency of micronucleated polychromatic erythrocytes (PCEs) in mouse bone marrow was assessed after administration of dipyridamole and/or adenosine monophosphate (AMP) to nonirradiated mice or to mice irradiated 15 min later with a sublethal dose of 6.5 Gy γ rays. In nonirradiated mice, the administration of the drugs increased the frequency of micronucleated PCEs significantly (by 108%). In contrast, in irradiated mice, the number of radiation-induced micronucleated PCEs was significantly decreased if the mice had been pretreated with dipyridamole or AMP alone (by 24% after administration of each of the compounds) and in particular after administration of the drugs in combination (by 36%). © 2000 by Radiation Research Society

INTRODUCTION

Adenosine nucleotides have been known for a relatively long time to exert radioprotective effects in mice when they are given shortly before irradiation (1, 2). Since the protective effects of these compounds are not dependent on the degree of their phosphorylation (3), and exogenously administered adenine nucleotides are rapidly metabolized extracellularly by ectonucleotidases (4), adenosine appears to be the ultimate effector of radioprotective action. We have shown in a number of experiments that the radioprotective effects of adenosine monophosphate (AMP) can be enhanced by dipyridamole. Dipyridamole is a potent inhibitor of nucleoside uptake by cells, and it elevates the extracellular concentrations of both endogenous and exogenous adenosine and prevents its intracellular metabolism (5). Our earlier experiments demonstrated the radioprotective effects

of combined treatment with dipyridamole and AMP in enhancing hematopoietic recovery and reducing the mortality of γ -irradiated mice (6–9). In addition, the combination of dipyridamole and AMP has been found to protect irradiated mice from early cellular damage, as shown in studies of the thymidine level in plasma and the amount of free polynucleotides in the thymus and spleen (10, 11). The protective effects of the joint administration of both drugs suggests that their action is based on the activation of extracellular adenosine cell receptors which can mediate various physiological effects (12, 13). At least two independent effects of the activation of adenosine receptors have been proposed to be responsible for the observed radioprotection (6–10): tissue hypoxia induced by the hypotensive action of adenosine, and enhancement of the proliferation in the compartments of hematopoietic cells.

The present study was undertaken to assess the effects of dipyridamole and AMP given alone or in combination as a protective regimen against radiation-induced DNA damage in terms of the frequency of chromosome fragments (micronuclei) in polychromatic erythrocytes (PCEs) in the bone marrow of mice. The *in vivo* micronucleus assay in mouse bone marrow provides an indirect measure of the induction of chromosome aberrations and is probably the most frequently used *in vivo* test of short-term genotoxicity (14–17).

MATERIAL AND METHODS

Animals

Three-month-old male (CBA $\times$ C57BL/10)F₁ mice weighing 28 g on average were used. Standardized pelleted diet and HCl-treated tap water (pH 2–3) were given *ad libitum*. The mice were kept under controlled lighting conditions (light:dark 12:12 h), and the temperature was maintained at $22 \pm 1^\circ\text{C}$. The research was conducted according to the principles enunciated in the Guide for the Care and Use of Laboratory Animals issued by the Czech Society for Laboratory Animal Science.

Drugs

Dipyridamole (Sigma, St. Louis, MO) was dissolved in 0.4% tartaric acid and injected subcutaneously (s.c.) at a dose of 2 mg per mouse in a

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TABLE 1
Effects of Dipyridamole and Adenosine Monophosphate in Nonirradiated and Irradiated Mice on the Indices of the Micronucleus Assay

	Number of micronucleated PCEs ^a per 1000 PCEs ^b	Number of PCEs per 100 (PCEs + NCEs ^c)
1. Untreated control	4.16 ± 0.69 ^{3,4,5}	55.44 ± 5.34
2. Tartaric acid ^d + saline ^e	4.31 ± 0.48 ^{3,4,5}	53.80 ± 5.45
3. Dipyridamole + saline ^e	5.45 ± 1.21 ^{1,2,4,5}	56.69 ± 4.92 ⁵
4. Tartaric acid ^d + AMP	8.15 ± 1.46 ^{1,2,3}	54.05 ± 6.02
5. Dipyridamole + AMP	8.67 ± 2.00 ^{1,2,3}	51.72 ± 4.06 ³
6. γ rays	60.18 ± 8.25 ^{8,9,10}	22.46 ± 3.42 ^{7,8,9,10}
7. Tartaric acid ^d + saline ^e + γ rays	54.28 ± 10.36 ^{9,10}	10.96 ± 2.63 ^{6,10}
8. Dipyridamole + saline ^e + γ rays	45.87 ± 10.37 ⁶	10.22 ± 1.96 ⁶
9. Tartaric acid ^d + AMP + γ rays	45.56 ± 5.38 ^{6,7,10}	9.77 ± 2.04 ⁶
10. Dipyridamole + AMP + γ rays	38.27 ± 5.08 ^{6,7,9}	8.88 ± 1.59 ^{6,7}

Notes. The data represent means ± SEM of values obtained from five mice. Irradiation significantly increased ($P < 0.01$) the numbers of micronucleated PCEs per 1000 PCEs and significantly decreased ($P < 0.01$) the numbers of PCEs per 100 (PCEs + NCEs) compared to nonirradiated mice in all groups of animals. The table shows only significant differences within the groups of nonirradiated or irradiated animals. The numbers¹⁻¹⁰ denote significance with respect to groups 1-10. For the sake of simplicity, only the level of $P < 0.05$ was used.

^a Micronucleated polychromatic erythrocytes.

^b Polychromatic erythrocytes.

^c Normochromatic erythrocytes.

^d Control solution for dipyridamole.

^e Control solution for AMP.

volume of 0.4 ml. Adenosine 5'-monophosphate sodium salt (AMP) from yeast (Sigma) was dissolved in distilled water and injected intraperitoneally (i.p.) at a dose of 5 mg free acid per mouse in a volume of 0.2 ml. Dipyridamole and AMP were given either alone or in combination; in the combination treatment, dipyridamole was administered 20 min before AMP. Tartaric acid (0.4%) and saline were used for control s.c. and i.p. injections. AMP or saline was administered 15 min before irradiation. The treatment regimen and doses of the drugs were those found to be radioprotective in our earlier experiments (6-11).

Irradiation

Mice were given a total-body dose of 6.5 Gy from a ⁶⁰Co γ-ray source at a dose rate of 0.29 Gy/min. During irradiation, the mice were placed in a ventilated Plexiglas container.

Bone Marrow Preparation and Staining

Mice were euthanized by cervical dislocation 24 h after irradiation or sham irradiation. Smears of femoral bone marrow were prepared and stained with May-Grünwald/Giemsa (14, 18). With this method, PCEs are stained reddish-blue and normochromatic erythrocytes (NCEs) are stained orange while nuclear material is stained dark purple.

Slide Analysis

The number of micronucleated PCEs among 1000 PCEs per mouse and the number of PCEs among 100 erythrocytes (PCEs + NCEs) were determined for each mouse.

Statistical Evaluation

Each experimental group consisted of five mice. Numbers of micronucleated PCEs and total PCEs were evaluated on two to five different slides from one bone marrow sample from each mouse. Analysis of variance did not reveal even marginally significant differences between the variability of the values obtained for the same mouse and the interindividual variability of the values within the experimental groups. Significance of differences in the effects of the treatment was analyzed by non-

parametric techniques (Kruskal-Wallis ANOVA and Mann-Whitney test) using the principles of Holm's (19) sequentially rejective multiple test procedure. In addition, nonparametric correlation and parametric approaches for comparison of the linear regression coefficients were used. The level of significance was set at $P < 0.05$.

RESULTS

A summary of the numbers of micronucleated PCEs per 1000 PCEs and PCEs per 100 (PCEs + NCEs) in the bone marrow samples from nonirradiated and irradiated mice treated in various ways is given in Table 1.

Micronucleated PCEs per 1000 PCEs

In nonirradiated mice, no significant difference was found between untreated controls and mice given the control solutions. Administration of dipyridamole increased the numbers of micronucleated PCEs moderately but significantly compared to these two control groups. Injection of AMP induced a greater enhancement of micronucleated PCEs, leading to values significantly higher not only compared to the two control groups but also compared to the mice given dipyridamole. A similar effect was induced by the combination of dipyridamole + AMP.

Irradiation induced a dramatic, highly significant, increase in micronucleated PCEs, as seen when comparing irradiated groups to nonirradiated groups (Table 1). While the yield of micronucleated PCEs was lower for the animals given control solutions before irradiation compared to those that were only irradiated, this reduction is not significant. When dipyridamole or AMP alone was administered before irradiation, a significant decrease in micronucleated PCEs

was observed compared to the mice that were irradiated only, and in the case of AMP also compared to the mice given the control solutions. After the administration of the combination of dipyridamole + AMP, a further decrease in micronucleated PCEs was seen; the value for micronucleated PCEs in this group was significantly lower not only compared to the mice that were only irradiated or the irradiated animals given the control solutions but also compared to the mice given AMP only before irradiation.

PCEs per 100 (PCEs + NCEs)

In nonirradiated mice, a significant difference in the number of PCEs per 100 (PCEs + NCEs) was found only between mice given dipyridamole alone (the highest numbers) and mice given the combination of dipyridamole + AMP (the lowest value). However, these effects do not appear to be substantial; the difference between the highest and the lowest value was less than 10%.

Irradiation induced a significant decrease in PCEs, as shown in a comparison of the irradiated groups with the nonirradiated groups. A significant difference in numbers of PCEs was found when the groups of mice that were irradiated only and mice irradiated and pretreated with the control solutions were compared. Administration of dipyridamole or AMP alone before irradiation did not decrease the values significantly compared to those for irradiated mice given only the control solutions. A further decrease to 81% of the value obtained in mice treated with control solutions was found in the group given the combination of dipyridamole + AMP; the difference between these two groups was significant.

DISCUSSION

The results of the present experiments clearly demonstrate the expected enhanced yield of micronuclei in bone marrow of irradiated mice (20, 21) and suggest a decrease in the occurrence of micronuclei caused by pretreatment with the drugs. The proliferative status of bone marrow erythroid cells must be taken into account in the interpretation of the effects of the drugs. Micronuclei can form only in dividing cells, and reduction of cell proliferation can decrease their yield (17). The numbers of PCEs (in relation to NCEs) are an indicator of the rate of proliferation, and their decrease 24 h after irradiation is an expression of the known early effects of radiation on the cell cycle. The radiation-induced decrease in PCEs was significantly enhanced by the injection of control solutions. Other carriers such as oils were also found to inhibit proliferation of bone marrow cells (22), and thus the action of nonspecific stress action can account for these effects. Glucocorticoids, which are mediators of nonspecific stress reaction, have been shown to suppress the mitotic activity of early erythroid progenitors (23) and thus can synergize the inhibitory effects of radiation on the erythroid system. In our experi-

ments, the effect of the control solutions was not observed in nonirradiated mice, probably due to the operation of compensatory proliferation mechanisms in the undisturbed populations of hematopoietic cells.

To ascertain the role of the bone marrow proliferative status in the occurrence of micronuclei in drug-treated and irradiated mice, a nonparametric and parametric evaluation of the relationship between the values of total PCEs and micronucleated PCEs was performed. When using the nonparametric technique in a sample consisting of all irradiated mice irrespective of the mode of treatment, the numbers of total PCEs were found to correlate significantly with the frequencies of micronucleated PCEs (Spearman test, $r = 0.48$, $P < 0.001$). However, the nonparametric approach cannot reveal the possible existence of differences in these relationships between various experimental subgroups. As shown in Fig. 1, two linear regressions characterizing the relationship between total PCEs and frequencies of micronucleated PCEs can be defined when using the parametric approach: (1) regression line based on values for control mice (intact and carrier-treated), exhibiting a shallow slope which we propose represents the measure of the "pure" effect of proliferative status on occurrence of micronuclei, and (2) regression line computed from values obtained in drug-treated mice (dipyridamole or AMP alone, dipyridamole and AMP in combination) and characterized by a steep slope. The slopes of these two regressions are significantly different at $P < 0.02$. Thus the results suggest that the decrease in micronucleus formation in drug-treated animals is due not only to the decrease in cell proliferation but also to the additive protective mechanisms of the drugs. Concerning the degree of protection obtained, the effects observed are comparable to those found in experiments with the potent radioprotector WR-2721 when using the same *in vivo* model (20, 21).

The reduction in the frequency of micronucleated PCEs observed in the variously treated groups of irradiated mice correlates approximately with the degree of transient systemic hypoxia culminating in the period 15 to 40 min after administration of the drugs as demonstrated in our previous experiments using measurements of blood pressure, heart rate and oxygen tension in murine tissues (7). The short-term reduction in blood pressure due to bradycardia and vasodilation are known effects of the receptor action of extracellular adenosine (24, 25). The greatest protection obtained with the combination of dipyridamole + AMP is compatible with the concept (see the Introduction) that both AMP, which serves as a source of exogenous adenosine, and dipyridamole, which inhibits adenosine uptake by cells and preserves it in the extracellular fluid, act jointly in enhancing the extracellular action of adenosine. Because the hypoxia-induced radioprotection is a classical radiobiological phenomenon, hypoxia induced by the action of adenosine seems to be the most reasonable explanation of the results. However, the role of other mechanisms cannot be excluded. For example, it has been shown that the addition

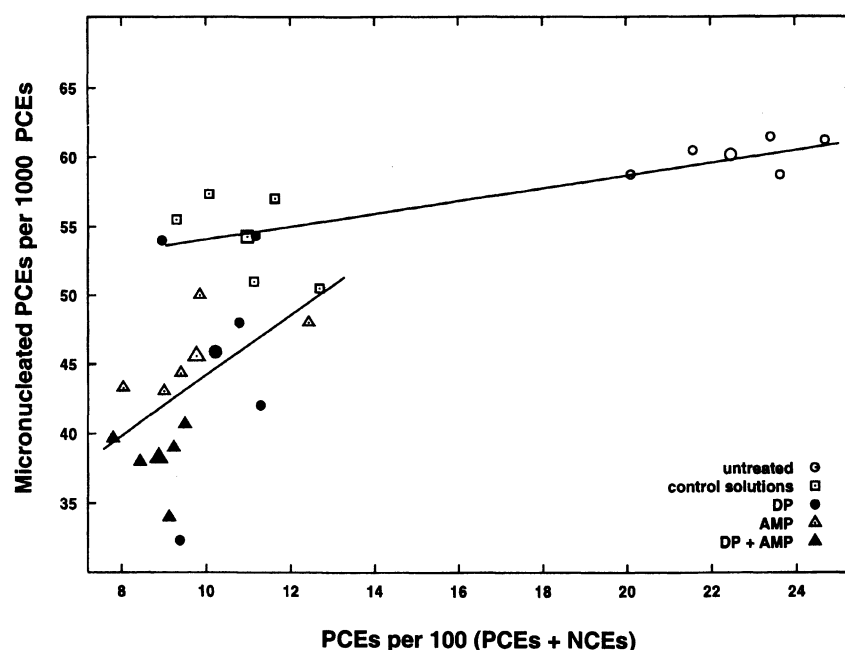


FIG. 1. Regression lines for the relationship between frequencies of micronucleated PCEs to total PCEs in irradiated intact mice, in mice treated with control solutions, and in irradiated mice pretreated with dipyridamole (DP) or adenosine monophosphate (AMP) alone and dipyridamole + AMP. Small symbols denote individual values; large symbols indicate group means.

of dipyridamole + AMP to *in vitro* irradiated thymocytes enhances the rejoining processes of DNA strand breaks (11). It is also known that dipyridamole scavenges oxygen radicals and inhibits lipid peroxidation (26).

Interestingly, significant increases in the number of micronucleated PCEs were found in nonirradiated animals treated with AMP alone and in combination with dipyridamole. The hypoxic effects of the drugs can also account for these findings. On the basis of radiobiological experiments, it was established that hypoxia not only protects cells from radiation damage through the "oxygen effect" but also inhibits mechanisms of intracellular repair (27, 28). It can be hypothesized that the reduction of the effectiveness of repair mechanisms under hypoxia in nonirradiated mice interferes with the repair of spontaneous DNA breaks and results in an increase in the frequency of spontaneous micronuclei. Similar effects were observed in experiments evaluating the action of thiol radioprotective compounds which decreased the micronucleus formation in irradiated mice but produced clastogenic effects in nonirradiated mice (20, 21). The decrease in oxygenation of tissue is a known effect that is also induced by thiol radioprotectors *in vivo* (29). The clastogenic effects of various radioprotective substances in normal nonirradiated organisms deserve further attention.

The findings of radioprotective effects of dipyridamole and/or AMP administered shortly before a sublethal radiation dose on the clastogenic effect of radiation measured as a decrease in the frequency of micronucleated PCEs extend our previous studies of the protective action of these drugs (6–11). The data presented here suggest that the drugs can

also decrease the mutagenic effects of ionizing radiation. The experimental evidence suggest that the use of dipyridamole and AMP, which are both available for clinical use, might be extended to conditions of acute radiation damage.

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